Personalized Emotion Detection from Text using Machine Learning

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***Abstract*—Social media platforms have evolved into essential tools for sharing emotions with the rest of the globe. People utilize text, photos, music, and video to express their emotions or their points of view. On the Internet, these social media platforms generate a vast volume of unstructured data. All the existing emotion detection models based on the supervised learning methodology require to collect labelled data which is time- consuming task. Next, for languages that are underutilized in terms of NLP, such as Polish vocabulary. To tackle these issues, this paper presents a model that functions as an emotional awareness system. It is based on open forums with automated audio information and text analysis for user-based emotion detection. The model is a key component of many natural computer programmers for processing natural language. The proposed model is implemented using PocketSphinx as an ASR. In addition to this, the model also uses the Word2Vec model, K-means clustering, and the Term frequency-inverse document Vectorizer for text analysis and segmentation. Further, the data from the International Survey on Emotion Antecedents and Reactions (ISEAR) was used to test and train the model. Finally, the concept evolves into a user-based system that will act as a personal assistant for segregating people's emotions as they experiment with new apps. The ISEAR database will be used for performance testing. When utilized in the database, the suggested model greatly outperforms previous models, with an accuracy of 80.25 percent and an F1 score accuracy of 88.86 percent. As the future work, this algorithm can be integrated into an application where emotion is used for recommendation.**

Keywords— PocketSphinx, Word2Vec, Automated Speech Recognizer, ISEAR, Emotion detection.

I. Introduction

Conversation is highly influenced by emotions. The emotions are also reflected in the context of the text. Recent years have seen an increase in interest in the task of emotion recognition from text. Recognizing emotions from text using machine learning now offers considerable and promising results due to the development of machine learning algorithms and hardware to enable the machine learning process. Recognizing emotions seems to be a simple task for a human, but it is a challenging task for the computer. It is possible to use machine learning to train the computer to recognize emotions in the context of a dialogue. The training data supplied as the fuel in the machine learning training should contain a lively conversation in order to provide excellent results. Social media is one of the potential dataset resources. People can share their disagreements, emotions, opinions, and tales in a lively and natural way on social media [1].

The models and classifiers cannot generalize when the training and testing environments are mismatched, which is one of the major issues in emotion recognition [2]. When data from an individual in the testing dataset is not included in the training dataset, this problem emerges. Speaker-dependent classifiers outperform speaker-independent classifiers, according to previous research [3]. This comment implies that the speaker's emotional expression is constrained. Although speakers share comparable patterns, they are insufficient to develop accurate emotion recognition algorithms [3]. The issue is that real-world operations need pre-built classifiers for unheard speakers' provocative speech. The research looks at how to acclimating an emotion recognition system to a specific user. For emotion identification, feature and/or model change is an appealing strategy. In the model adaptation frame, using data with emotional labels from the user is a challenge, which limits its applicability in many practical procedures [4]. This work offers an unsupervised approach to acclimate an emotion recognition system to a given user in this setting. Manually labelled data is required for supervised learning, which is time-consuming and not always practicable. Traditional styles have a low level of accuracy in the SER (Speech Emotion Recognition) system. There is a need for a validated emotion model to make the recognition process more realistic since different events create different emotions in different persons, which can be addressed effectively by it. Keeping in mind all the issues discussed above, we propose a hybrid approach to solve this concern. The proposed approach intends to reduce the disparity between auditory characteristics used during training and those derived from the targeted speaker during testing. The ISEAR (International Survey on Emotion Antecedents and Responses) dataset [5] is used to create a baseline emotion identification algorithm. Videos from a popular website for videotape participants are used to "retrieve 1.5 hours of speech from the targeted speaker."

All the existing emotion detection models based on the supervised learning methodology require to collect labelled data and train a supervised algorithm to understand each word in a sequence that relates to the overall sentence's negative or positive outcome. This kind of method requires data to be labelled by humans, which takes time and isn't always possible. Secondly, for languages that are underutilized in terms of NLP, such as Polish vocabulary. There are not as much trained models to work with; moreover, the libraries that have previously pretrained models for predicting sentiment ratings for each word in a sentence are not always available. To tackle all above issues, a hybrid model is proposed in this paper that uses PocketSphinx as an ASR. The model also uses the Word2Vec model, K-means clustering, and the Term

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frequency-inverse document Vectorizer for text analysis and segmentation.

The paper is organized as follows: The literature review is mentioned in Section II. The proposed model's architecture is presented in Section III. Section IV covers the dataset, whereas Sections V and VI discuss the implementation and the results, respectively. Finally, the conclusions and future work are described in Section VII.

II. Literature Review

An emotion-based integrated architecture [6] is employed with semantic analysis to extract semantic information and create ontologies based on emotion models. Further, case-based reasoning [7] is used to adopt new terminology and applications. The data is matched to each case stored in the case storage. The difference between the data and each case is calculated using algorithms to determine the degree of similarity. The estimating method was not specified, moreover, the input data and case quality were also not determined, all of which were done manually, resulting in reduced accuracy.

Semantic analysis employs natural language processing techniques such as statistic-based parsing and dependency trees to obtain more information about phrases. The use of verbal information to extract keywords from sentences has been suggested [8].

The models adopt the foundational science that allows us to connect the results of semantic analysis to believable feelings. The OCC model, which includes 22 different emotion types, is used. A more systematic technique for analyzing many textual sources evolved when emotion models and semantic analysis were integrated [9].

A fusion-based architecture consists of a keyword-based component and a knowledge system component [10]. The sentence splitter, which is largely based on the knowledge engineering technique for information extraction, can help in finding hidden word patterns and improving the contextual analysis. By using a tokenizer and POS tagger, information can be received about semantic and syntactic. The learning component builds feature vectors and tests prediction accuracy using the LibSVM format. Machine learning can be broken down into two parts. The first step is used to create an emotion prediction training model. In the second step, the learned model is then used to predict the sentiment class using a testing dataset. They however, perform a large amount of domain-specific bias division.

The purpose of normalization is to minimize speaker inconsistency while preserving emotional class discernment. This was achieved by minimizing the variances in audio features utilized to train and evaluate the emotion identification system [11]. They used iterative feature normalization (IFN) as a front-end method. But in spite of using all the above methods, some samples were still misclassified, according to IFN.

The authors in [12] performed a complex syntactic and semantic analysis of the text and used multiple ontologies such as Wordnet and Concept Net during emotion recognition process. It results in improved coverage of emotion rules.

In [13], emotion identification was carried out using a keyword-based strategy that primarily targeted phrasal verbs. They employed the keyword-based technique after preprocessing the ISEAR [14] data. They constructed their own database after identifying various phrasal verbs that should have been connected to emotion categories but weren't. They identified phrasal verbs and terms that were interchangeable with different emotions and classified them by utilizing their database. However, while achieving a considerably greater accuracy of 65%, they were unable to resolve the researcher's preexisting problems, such as a dearth of emotion keywords and a disregard for word semantics in interpretation.

Based on the detection technique, text-based emotion detection can be divided into three groups [15].

A. Approach Based on Keywords

At the word level, keyword-based techniques are applied. They need an emotion keyword dictionary, which covers terms and emotion labels. This method is used to come up provide a list of criteria for building a language framework that can be used to assess emotion [15].

B. Learning-based Technique

In this technique, the goal of distinguishing sentiment from text is defined as a set of input texts with sentiments as classes. In the supervised method, datasets that have been manually described are used to train a classifier, which is subsequently applied to other groups. In unsupervised procedures, annotated training data is not required. These approaches rely on lexical resources to measure the mood or polarity of words, sentences, and documents. A corpus is a large, well-organized collection of texts that are all related to the same subject or author [16].

C. Fusion Method

The fusion method [15-16] is a hybrid method that combines corpus and knowledge-based methodologies. It is divided into two types based on the viewpoint of the user: A single text fragment could provoke a wide range of emotions in the reader.

From the writer's perspective, one paragraph section represents only one emotion [17]. The authors of [16] used a mix of natural language processing techniques and machine learning to estimate the influence in the form of six main emotions. The algorithms based on deep learning are made up of a number of nonlinear, simultaneous components overcoming the constraints of existing techniques. These approaches must be constructed with deeper levels of framework than present methodologies.

For multiclass text analysis and classification [18], the authors employed Pocket-sphinx as an Automatic Speech Recognizer, Human Computer Interaction (HCI), and linear support vector machine (Linear SVM).

III. Proposed Model and Methodology

We offer a customized emotion detection approach in order to make the recognition process more realistic. The flow of the model is determined by the input. If the input is in the form of speech signals, then it attempts to classify human emotions that is extracted out of Automated Speech Recognizer (ASR) along with text processing, but, if the input is text, then it can be directly classified from this model.

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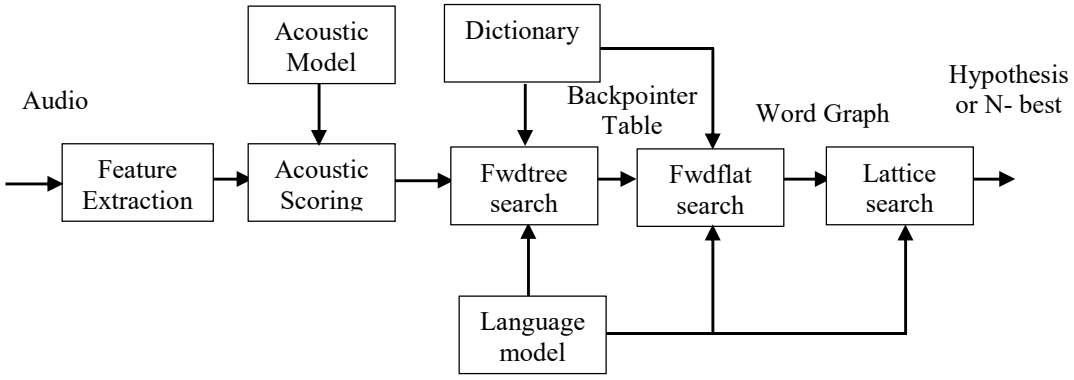

Fig.1 PocketSphinx [18]

The proposed model uses PocketSphinx for ASR and a Word2Vec based model for classification. Numerous recognizers were compared with a standard database in prior studies, and another study analyzed the real-time component, i.e., PocketSphinx. The PocketSphinx ASR takes a wav audio file and converts it to Mel frequencies cepstral coefficient feature sets (MFCC). To map and generate the specific text output, this MFCC is used in a knowledge base that includes a phonetic dictionary, language model, and acoustic models.

The various modules of PocketSphinx are shown in Fig. 1. It is used for mapping of vocabulary words to phoneme sequences that is provided by a phonetic dictionary. In the absence of a dictionary, the recognizer won't be able to recognize the words that is been looking for. Even so, having the terms in the dictionary is not enough. The recognizer searches the dictionary and the language model for a word. A term will not be recognized without the language model, even if it is in the dictionary.

The language model is the key part of the design that tells the decoder about sequences of words that are possible to recognize. There are several types of models' keyword lists, grammars and statistical vocabulary models, and phonetic language models. They have diverse capabilities and performance properties. One can opt any decoding mode corresponding to one's requirements and can indeed switch between modes in runtime. It is a statistical model based on the Hidden Markov Model. A Gaussian distribution describes the HMM output of the states once again. Acoustic Adaption techniques in Maximum Likelihood linear Regression (MLLR) and PocketSphinx include Maximum-APosteriori (MAP). Despite the fact that both MAP and MLLR can construct speaker-dependent systems, we find that they are currently incompatible. As data size expands, MAP adaptability overtakes MLLR.

$$\text{MAP} = \text{arg maximum } f(x)*g() \tag{1}$$

The MAP is defined in equation (1) as where f(x) denotes the probability and g() denotes the former distribution.

The Word2Vec uses a neural network model to learn word connections from a large corpus of text. After being educated, it can recognize synonyms. A vector, which is a collection of figures, is used to represent each word. The continuous skip-gram and Continuous Bag-of-Words (CBOW) [11] are two models that Word2Vec can employ. In the continuous bag-of-words architecture, the model guesses the current word from a window of surrounding environment words. The arrangement of the environment terms has no bearing on the outcome (bag-of-words assumption). Regarding the existing word, the continuous skip-gram architecture predicts the surrounding window of environment words.

The neural network is programmed to achieve this by-passing word pairs to it from our training articles. Fig. 2 shows some of the training samples (word pairs) taken from the sentence "The quick brown fox jumps over the lazy dog". The window size is 2. The word emphasized in blue is the input word.

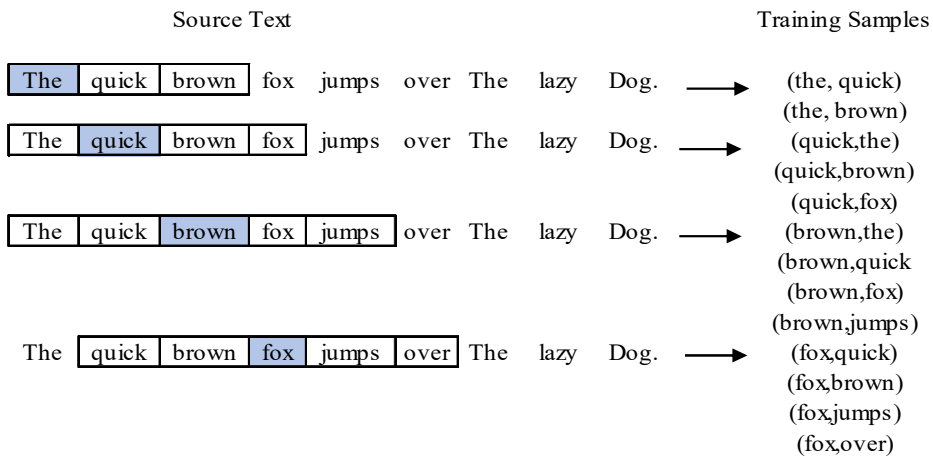

Fig. 2 Word2Vec [12]

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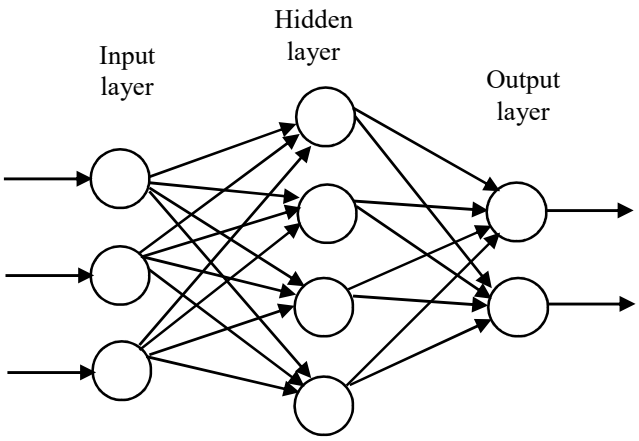

Fig. 3 Neural network [2]

The network will explain the chance of each word in our vocabulary having the "proximate term" that has been chosen. The neural network is trained by inputting it with word pairs recognized in the training documents. The architecture of neural network is shown in Fig. 3.

As a word cannot be inputted to a neural network as a text string, therefore, a way is needed to communicate the words to the neural network. Accordingly, we compile a vocabulary of terms from training materials–let's say there are 10,000 distinct words. Then, using an input word like "books," a single-hot vector is constructed. This vector will have around 20,000 components (one for each word in our vocabulary), with a "1" in the location corresponding to the phrase "books" and a "0" in all other locations. The network produces a single vector (again, with 20,000 components) with a single component for each word in our vocabulary: the probability that a randomly picked nearby word is that vocabulary word.

In the proposed model, the fastest PocketSphinx speech recognition system is used as a real-time speech translator. PocketSphinx takes speech input in the form of a waveform and separates it into utterances, such as a cough, "oh," or breath. PocketSphinx attempts to match every conceivable word combination with the audio. Then, it selects the most matched pairing. It is also used for noise suppression. Thereafter, the Word2Vec framework is employed, which uses the preceding and following context to broadly understand the meaning. The full sentence of the data is trained using the Word2Vec model to vectorize the words in order to grasp the meaning according to the sequence of the words. Further, K-means clustering is performed on word vectors obtained in the earlier stage and the values are assigned based on the cluster to which they belong. The weighted sentiment coefficient was then computed by multiplying the values by their proximity to the cluster. Each sentence now contains two vectors: a weighted sentiment score and a tfidf score. The emotion identified is then determined by finding the dot product of the two vectors for each sentence and calculating the tfidf score. K-means is utilized because it can handle big datasets, always converges, and adapts to new examples. Tfidf is employed because a value is given to each term. The proposed model is shown in Fig. 4.

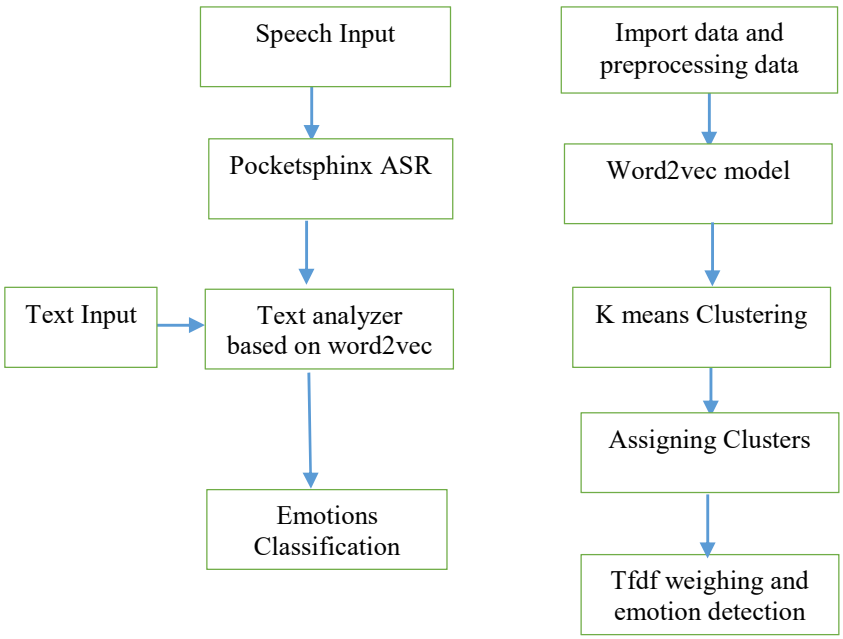

Fig. 4. Architecture of proposed model

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IV. Dataset

The proposed architecture is evaluated using ISEAR (International Survey on Emotion Antecedents and Reaction) [19]. As per ISEAR, there are seven primary emotion classifications. Among them are wrath, contempt, fear, shame, pleasure, embarrassment, and sadness. There is a classification label for each sentence. Klaus R. Scherer and Harald Walcott led the ISEAR study [19], which gathered information from a varied group of psychologists across the world. The students, psychologists and non-psychologists, were asked to identify situations in which they had experienced all seven primary emotions (wrath, contempt, fear, shame, pleasure, embarrassment, and sadness). The questions in each example focused on how they viewed the situation and responded. As a result, the final data collection included reports on seven sensations from almost 3000 respondents from 37 countries across five continents.

V. Implementation

In the proposed model, the dataset used is ISEAR. The Genism's implementation of the word2Vec algorithm with skip gram architecture is used in the proposed model. The sklearn implementation of the K-means method is employed with 50 repeated beginning points and 1000 iterations of re-assigning points to clusters. The learning rate is dropped from 0.03 to 0.0007 using a lookup window of 6 and a negative sampling of 20 words. The score is multiplied by how near they are to their cluster to determine its weight. Following these steps, a complete dictionary is created (in the form of a Pandas Data Frame), with each word having its own weighted sentiment score. The TF-IDF score of each word in each phrase is calculated using the Tfidf Vectorizer in sklearn. This step is conducted to see in each phrase, and to check disparity among words. The dot product of two phrase vectors (tfidf) and weighted sentiment scores revealed whether the overall mood is positive or negative and accordingly the feeling felt is positive or negative.

TABLE I. TABLE ON VARIOUS PERFORMANCE MEASURES

Performance Measure	Word2Vec Model
Accuracy	0.802503
Precision	0.99498
Recall	0.799930
F1-Score	0.888608

VI. Result analysis

This paper presents an unsupervised and tailored emotion detection framework to predict human emotion. The proposed emotion recognizer is based on a combination of ASR and NLP, is being trained and tested using the ISEAR database. The summary of results is described in Table I.

A customized emotion detection architecture is built that is used to categorize emotions. This model receives a user's text as an input, which is then passed on to the trained model, which calculates the anticipated emotion. Table I shows an example of the obtained output. Here, the obtained prediction accuracy is 80% using the word2Vec algorithm which is 10% more than the model used in [12] which uses an unsupervised feature adaption scheme. The obtained accuracy tells that word2Vec when used with the adequate parameters as stated earlier yields the best results. Our model ended up with a precision of 0.99, indicating that it is particularly good at recognizing negative sentiment observations. The model also had a F1-score of 0.88, which is one of the top outcomes, and an 80 percent recall, which means that 80 percent of all positive observations in the dataset were accurately identified as such. It is quite excellent at detecting negative sentiment observations. By the results obtained above, we believe that, as the dataset gets acquainted with more users' data, the emotion classification from the text can be extended to all of the users' emotion.

Based on the results, it is found that the emotion categorization from text to be expanded to cover all of the users' emotions as the dataset becomes stronger with more user data. In the future, we want to integrate it in a way that is interactive and can also be used as a personal assistant. The emotions that are gathered will be used to drive the recommendation system, which will make the process of making recommendations more specific

VII. Conclusions and future work

In the existing emotion detection systems, there are not as many trained models to deal with language. They are underutilized in terms of NLP, such as Polish vocabulary, so it is not practical to employ libraries that have previously trained models for predicting sentiment scores for each word in a sentence. Moreover, supervised- based emotion detection models require data to be labelled by humans, which takes time and isn't always possible. To tackle these issues, this paper presents an emotion detection model by combining ASR and text analysis. In the proposed model, the open-source tool Pocket-sphinx based ASR is used for speech to text conversion, and Word2Vec based Text Analyzer is used. It includes the global repository of emotion data. Further, algorithms such as K-means and TfIdf vectorizer is used. The input for our model is the user's text, and then, based on the text, the user's emotion is predicted. The testing showed that the proposed system has an accuracy of 80%.

References

[1] A. Chowanda, R. Sutoyo, and S. Tanachutiwat, "Exploring text-based emotions recognition machine learning techniques on social media conversation", Procedia Computer Science, vol. 179, pp. 821-828, 2021.

[2] T. Rahman, and C. Busso, "A personalized emotion recognition system using an unsupervised feature adaptation scheme," in 2012 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), pp. 5117-5120. IEEE, 2012.

[3] C.R. Aydin, and T. Güngör, "Combination of recursive and recurrent neural networks for aspect-based sentiment analysis using inter-aspect relations," IEEE Access, vol 8, pp. 77820-77832, 2020.

[4] K. Sridhar, and C. Busso, Unsupervised personalization of an emotion recognition system: The unique properties of the externalization of valence in speech, arXiv preprint arXiv:2201.07876 , 2022.

[5] A.A. Alnuaim, M. Zakariah, P.K. Shukla, A. Alhadlaq, W.A. Hatamleh, H. Tarazi, R. Sureshbabu, and R. Ratna, "Human-computer interaction for recognizing speech emotions using multilayer perceptron classifier," Journal of Healthcare Engineering, 2022.

[6] S. Munoz, O. Araque, J.F. Sánchez-Rada, and C.A. Iglesias, "An emotion aware task automation architecture based on semantic technologies for smart offices," Sensors, vol. 5, no. 18, pp. 1499, 2018.

[7] F. Lu, W. Si, and G. Tian, "Autonomous cognition and correction system of robot service based on emotional information and case-based reasoning," in IEEE 33rd Chinese Control and Decision Conference (CCDC), pp. 3235-3240, 2021.

[8] D.H. Maulud, S.R. Zeebaree, K. Jacksi, M.A. Sadeeq, and K.H. Sharif, "State of art for semantic analysis of natural language processing," Qubahan Academic Journal, vol. 2, no. 1, pp. 21-28, 2021.

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[9] G.L. Clore, and A. Ortony, "Psychological construction in the OCC model of emotion," Emotion Review, vol. 4, no. 5, pp. 335-343, 2013.

[10] A. Agrawal, and N.K. Mishra, “Fusion based emotion recognition system,” in IEEE 2016 International Conference on Computational Science and Computational Intelligence (CSCI), pp. 727-732, 2016.

[11] C. Lu, Y. Zong, W. Zheng, Y. Li, C. Tang, and B.W. Schuller, "Domain invariant feature learning for speaker-independent speech emotion recognition," IEEE/ACM Transactions on Audio, Speech, and Language Processing, vol. 30, pp. 2217-2230, 2022.

[12] N. Alswaidan, and M.E.B. Menai, "A survey of state-of-the-art approaches for emotion recognition in text," Knowledge and Information Systems, vol. 8, no. 62, pp. 2937-2987, 2020.

[13] D. Seal, U.K. Roy, and R. Basak, Sentence-level emotion detection from text based on semantic rules, in Information and Communication Technology for Sustainable Development, Springer, Singapore, 2020, pp. 423-430.

[14] A. Iliev, and P. Stanchev, "Information retrieval and recommendation using emotion from speech signals," in 2018 IEEE Conference on Multimedia Information Processing and Retrieval (MIPR), pp. 222-225, 2018.

[15] S. Al-Saqqa, H. Abdel-Nabi, and A. Awajan, "A survey of textual emotion detection," in IEEE 8th International Conference on Computer Science and Information Technology (CSIT), pp. 136-142, 2018.

[16] P. Nandwani, and R. Verma, "A review on sentiment analysis and emotion detection from text," Social Network Analysis and Mining, vol. 1, no. 11, pp. 1-19, 2021.

[17] R.A. Srivastava, and G. Deepak, PIREN: prediction of intermediary readers’ emotion from news-articles, in Data Science and Security, Springer, Singapore, 2021, pp. 122-130.

[18] F. Pantazoglou, N. Papadakis, and G. Kladis, Implementation of the generic greek model for cmu sphinx speech recognition toolkit, Proceedings of eRA-12, 2017.

[19] K.R. Scherer, and H.G. Wallbott, "Evidence for universality and cultural variation of differential emotion response patterning," Journal of Personality and Social Psychology, vol. 2, no. 66, pp. 310, 1994.

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